

MERT KOŞAN

PH.D. CANDIDATE

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WORK EXPERIENCE

Ph.D. Intern Data Scientist - Visa Research

Peer Group Analysis with Anomaly Detection

June 2021 – September 2021

Austin, USA

- Proposed a scalable and parallelizable dynamic peer grouping algorithm that works with any existing anomaly detection algorithm to reduce false alarms and detect anomalies in their early stages.
- Showed that the framework reduces the false positives substantially and detects multiple events earlier.
- Aiming to file a patent and possible extension to a paper.

Ph.D. Intern Data Scientist - Visa Research

Fraud Detection and Profiling

June 2020 – August 2020

Austin, USA

- Developed a framework that generates profiling and optimized strategies for near-real time unsupervised fraud detection on graphs. The proposed framework profiles the fraud automatically and accurately in less than a second with a small number of transactions.
- Filed a patent, which is currently under review.

Part-time Software Engineer - Proente

Industry 4.0 Solutions at Factories

June 2017 – August 2018

Istanbul, Turkey

- Developed a module generating statistics on data collected by Prowmes.

Intern Software Engineer - Robert Bosch GmbH

Automotive Industry

June 2016 – August 2016

Plochingen, Germany

- Designed a module generating reports on data collected by ESI [Tronic] 2.0.

TEACHING EXPERIENCE

Co-designer/Instructor - UC Santa Barbara

Machine Learning Workshops / Python

May 2019 – June 2019

Santa Barbara, USA

- Helped the students, from A Self-Organized PhD Students Exchange Program by Junior Nanotech Network, for their projects.

Teaching Assistant - UC Santa Barbara

Data Structures & Algorithms / C++, Python

September 2018 – April 2019

Santa Barbara, USA

Co-designer/Instructor - Sabanci University

Enhancement Workshops for Computer Science / Git, Linux, Python etc.

January 2018 – June 2018

Istanbul, Turkey

Teaching Assistant - Sabanci University

Database Systems & Operating Systems / MySQL, C++, Java

February 2016 – May 2017

Istanbul, Turkey

SKILLS

- Programming:** Python, C++, Java, Javascript, SQL
- Tools:** Neo4j, Git, LaTeX, Office, MySQL
- Languages:** Turkish (native), English (fluent), Spanish (beginner)
- Personality:** Result-oriented | Responsible | Hardworking
- Hobbies:** Football | Strategy Games | Chess | Snooker

EDUCATION

Ph.D. in Computer Science

University of California at Santa Barbara

2018 – 2023 (Expected)

Santa Barbara, USA

Advisor: Prof. Ambuj K. Singh | GPA: 4.00

Research Area: **Network Science, Data Mining, Machine Learning**

B.Sc. in Computer Science

Sabanci University

2013 – 2018

Istanbul, Turkey

GPA: 3.99 | Ranked 1st

SELECTED PROJECTS

Subgraph Detection using Bilevel Optimization

In Preparation, 2021

- Combining Graph Neural Networks and bilevel optimization to detect important subgraphs in the graph.

Black Box Modeling in Decision Making

In Preparation, 2021

- Analyzing how the risky behavior of individuals alters in the group setting from the individual setting using a black-box model comparing white-box models (e.g. Prospect Theory).

Event Detection on Dynamic Graphs

Under Review, 2021

- Proposed an event detection framework using Dynamic Graph Neural Networks further enhanced by self-attention. Tested on multiple real datasets.

Bike Sharing

September 2018 – December 2018

- Designed and implemented a bike sharing rebalancing problem using a network flow approach.

Impact Measurement of Tweets

October 2017 – June 2018

- Measured the hidden impact of a tweet by detecting copy-pasted or referenced tweets named as “Hidden Retweets” by Dr. Arin.

Differential Privacy Framework for Movie Recommendation System

September 2016 – September 2017

- Proposed and implemented a movie recommendation system that utilize differential privacy framework which guarantees the privacy of the individuals without losing the utility.

CryCur

Command line tool for crypto-currency applications

January 2017 – May 2017

Economics and Terrorism

Correlation of economic prospects (e.g. GDP) and terror incidents

January 2017 – May 2017